User Interface & User Experience



PeTIK Jombang

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12. Layout Windows & Halaman, Feedback, Guidance, dan Assistance yang efektif

UI/UX

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Layout Windows

UI/UX

Tujuan

- Encourage quick and correct information comprehension
- Fastest possible execution of tasks and functions
- Enhanced user acceptance

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Organizing and Laying Out Screens

UI/UX

Pendahuluan

- How a screen is organized and how its information is actually presented are crucial to achieving the design goals of fast and accurate comprehension and control execution.
- Following is a summary of numerous design principles that can be applied toward these ends.

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General Guidelines

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General Guidelines (1)

- Amount of information
  - Present the proper amount of information on each screen
  - Present all information necessary for performing an action or making a decision on one screen, whenever possible
- Organization and flow
  - Divide information into units that are logical, meaningful, and sensible.
- Organize by the degree interrelationship between data or information.

General Guidelines (2)

- Control placement
  - Position the most important and frequently used controls at the top left
  - Maintain a top-to-bottom, left-to-right flow.
- Navigation
  - The flow of interaction should:
    - Require as little cursor and pointer travel as possible
    - Minimize the number of times a person's hand has to travel between the keyboard and the mouse.
  - Assist users in navigating through a screen by: Aligning element, grouping element, using line borders

General Guidelines (3)

- Aesthetic:
  - Provide a visually pleasing composition through: Adequate use of white space, Balance, Groupings, Alignment of elements
- Visual clutter
  - Avoid visual clutter by: Maintaining low screen density levels, Maintaining distinctiveness of elements

General Guidelines (4)

- Focus and emphasis
  - Provide visual emphasis to the most important screen elements, its data or information
  - Sequentially, direct attention to items that are: Critical, Important, Secondary, Peripheral
- Consistency
  - Provide consistency
    - With a person's experiences and cultural conventions.
    - Internally within a system, including: Operational and navigational procedures, Visual identity or theme, Component, Organization, Presentation, Usage, Locations
    - Externally across systems

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Organization Guidelines

UI/UX

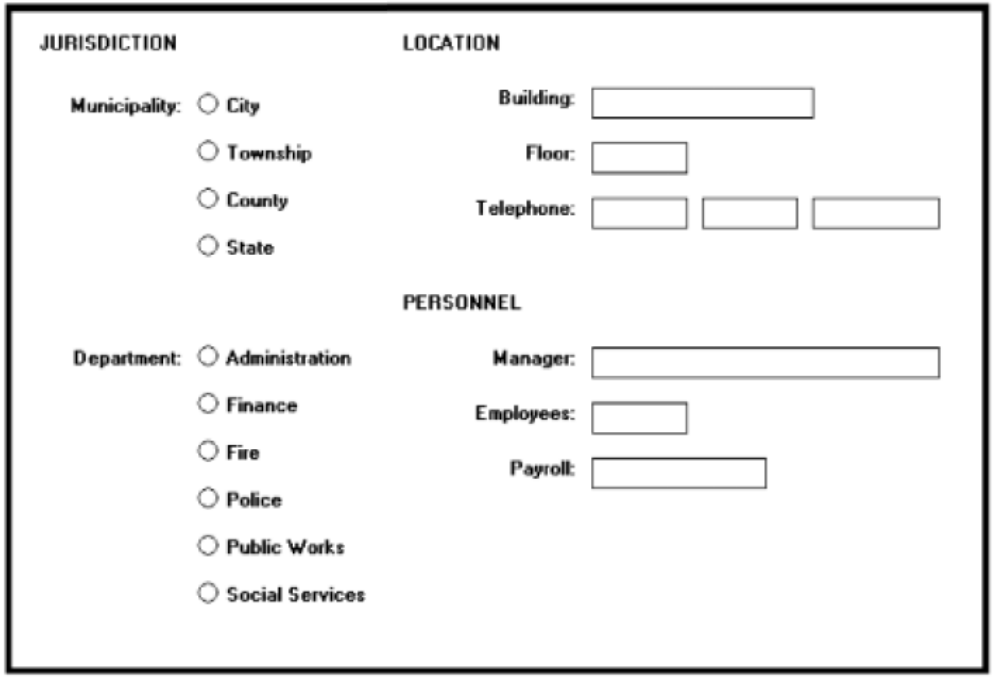
Organization Guidelines

Organization guidelines to be addressed include those relating to groupings, borders, dependent controls, alignment, and balance.

Grouping

- General: Provide groupings of associated elements
- White Space: Provide adequate separation of groupings through the liberal use of white space
- Headings: Provide section headings and subsection headings for multiple control groupings.
- Borders: Enhance groupings through incorporation of borders

Grouping

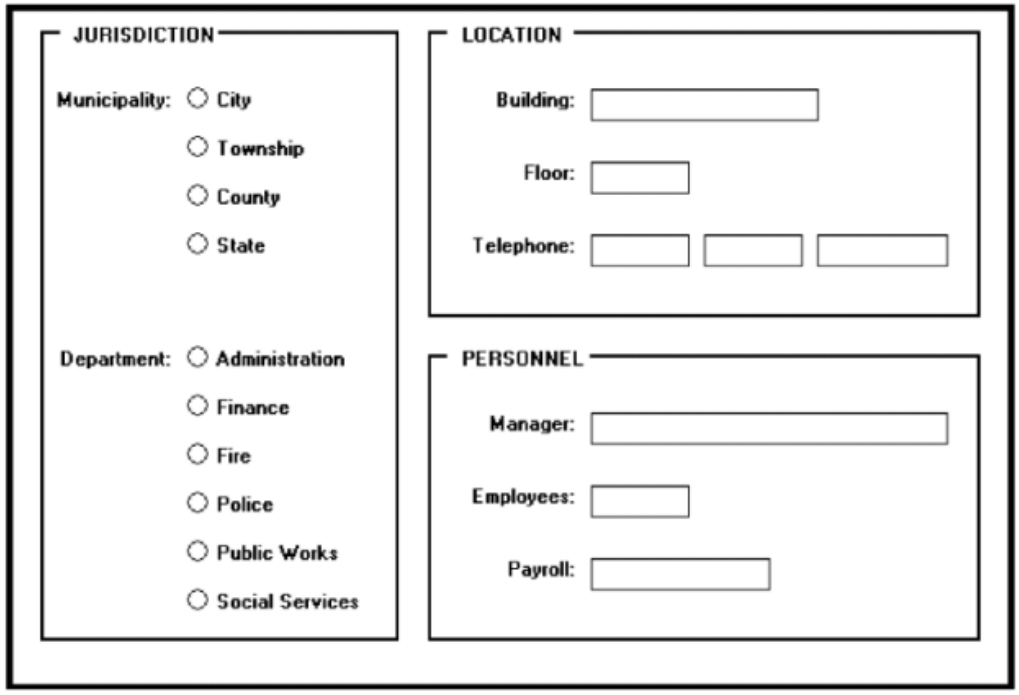


Grouping

Borders

- Incorporate a thicker single-line border around groups of related entry or selection controls.
- Locate the section heading in the top border, indented two character positions from the left border

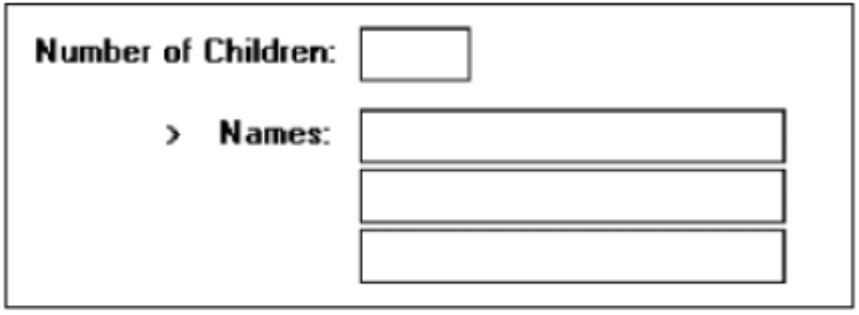
Borders



Grouping and Border

Dependent Control

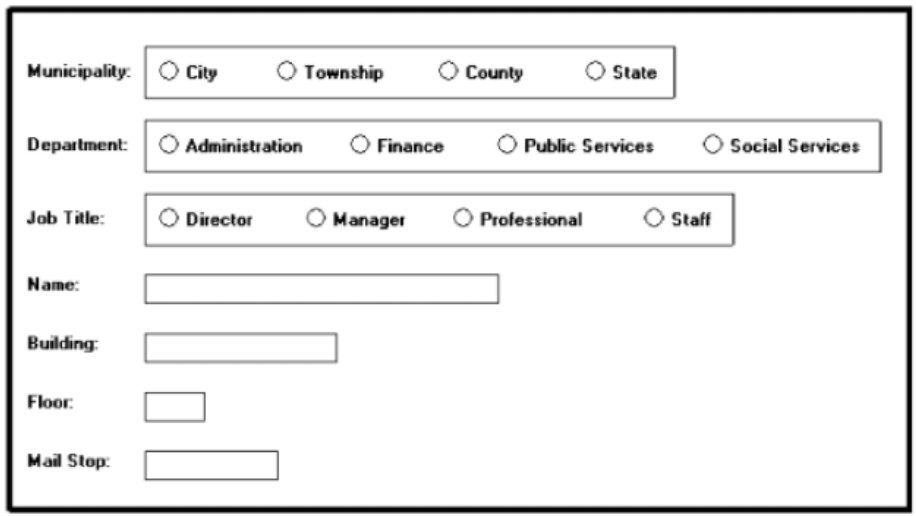
- Position a conditional control, or controls
- Inscribe a filled-in arrow between the selected control and its dependent controls to visually relate them to each other.



Dependant Control

Aligning

- Minimize alignment points on a window vertically and horizontally

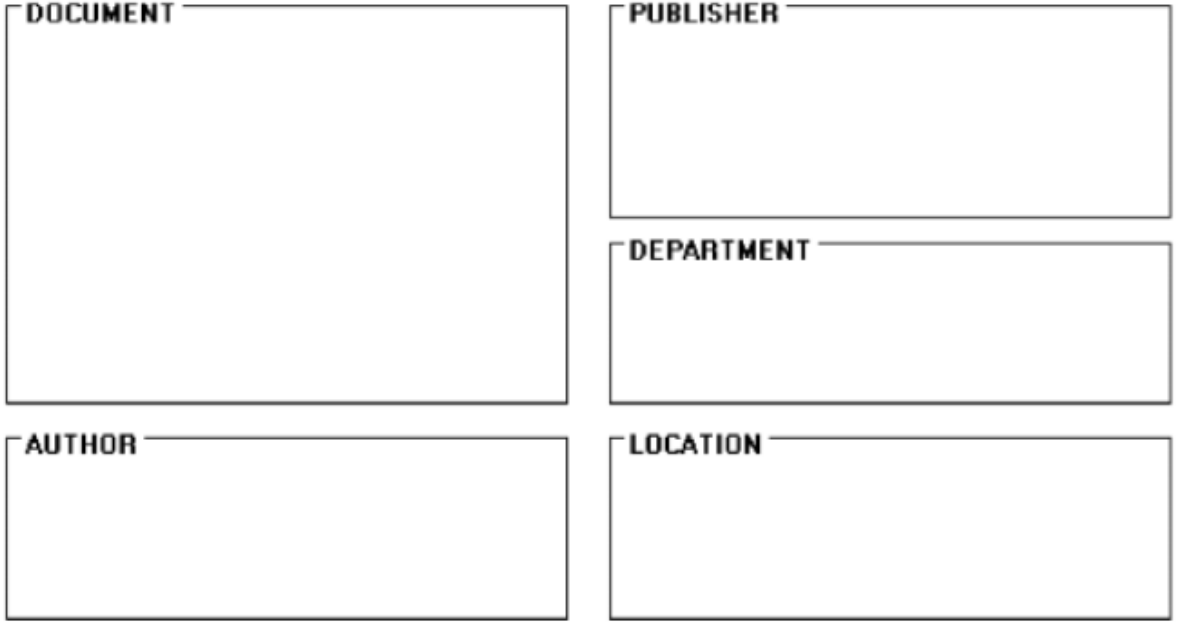


Alignment Vertically

Balance

- Create balance by
  - Equally distributing controls, spatially, within a window.
  - Aligning borders whenever possible.
- Individual control borders
  - Align the controls following the guidelines for multiple-control alignment
  - Align the left and right borders of all groups
- Section borders
  - Align the left and right borders of all groups
  - Establish the left and right border positions

Balance



Balance

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Windows Guidelines

UI/UX

Window Guidelines

- Organization
  - Organize windows to support task
  - Present related information in a single window
  - Support the most common tasks in the most efficient sequence of steps.
  - Use primary windows to begin an interaction and Perform a major interaction
  - Use secondary windows to Extend the interaction and display supplemental information
  - Use Dialog boxes for needed information and "Nice-to-know" information
- Minimize the number of windows needed to accomplish an objective

Window Guidelines

- Size
  - Provide large enough windows to
    - Present all relevant and expected information for the task
    - Prevent hiding important information.
    - Prevent crowding or visual confusion.
    - Minimize the need for scrolling.
    - Occupy less than the full size of the entire screen.
  - If a window is too large, determine is all the information needed or related

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Provide Effective Feedback

UI/UX

Feedback

- All user actions must be reacted to in some way
- Important to provide thorough and timely guidance and assistance

Feedback Topics

Feedback topics:

- Acceptable response times
- Dealing with time delays
- Blinking for attention
- The use of sound

Response Times

- System responsiveness should match the speed and flow of human thought processes
  - Response time less than 2 seconds if continuity of thinking is required and information must be remembered
  - Response time of 2 to 4 seconds if human task closures exist, high levels of concentration are not necessary
  - Responses within 4 to 15 seconds if major task closures exist, minimal short-term memory requirements are imposed
  - Response greater than 5 seconds if user is free to do other things and return when convenient,
- Constant delays are preferable to variable delays

Dealing with time delays

- Acknowledge all button clicks by visual feedback
- Present a “busy” signal If an operation takes 10 seconds or less
- Display A rolling barber’s pole If an operation takes longer than 10 seconds
- Display a progress indicator if over 1 minute
- Present an acknowledgment that it is completed when Long and invisible operations:
- Give the user something for pages requiring download times greater than 5 seconds

Blinking for attention

- Flashing an indicator when an application is inactive but must display a message
- Provide an auditory signal (one or two beeps)
- Display the message when application is activated and when requested by user

The use of sound

- Use in conjunction with a visual indication
- No more than six different tones.
- Do not use jingles and loud signals
- Use tones consistently
- Provide signal frequencies between 500 and 1,000 Hz
- Allow the user to adjust the volume
- Test the sounds with users over extended trial periods
- Use sounds sparingly

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Guidance and Assistance

UI/UX

Preventing Errors

- When errors occur, they must be properly managed
- Errors interrupt planning and cause deep frustration
- Errors can be classified as slips or mistakes
- A slip is automatic behavior gone awry.
- Slips can be reduced through proper application of human factors in design

Preventing Errors

- A mistake results from forming a wrong model and then acting on it.
- A mistake may not be easily detected
- Mistakes can be reduced, however, by eliminating ambiguity from design
- Doing usability testing and watching for nonsensical requests
- People should be able to correct them as soon as they are detected, as simply and easily as they were made.

Problem Management

- Prevention
- Detection
- Correction

Prevention

- Disable inapplicable choices.
- Use selection instead of entry controls
- Use aided entry
- Accept common misspellings, whenever possible
- Permit it to be reviewed before an action is performed
- Provide a common action mechanism
- Force confirmation of destructive actions
- Provide an automatic and continuous Save function

Detection

- For conversational dialog, validate entries (character, control, transaction)
- For high speed, head-down data entry
- Leave window open
- Maintain the item in error on the screen
- Visually highlight the item in error
- Display an error message in a window
- Handle errors as gracefully as possible
- Use auditory signals conservatively.

Correction

- Preserve as much of the user's work as possible
- At window-level validation, use a modeless dialog box to display an error list
- Always give a person something to do when an error occurs
- Provide a constructive correction message
- Initiate a clarification dialog if necessary.

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Providing Guidance and Assistance

UI/UX

Providing Guidance and Assistance

Providing Guidance :

- Form of the system's hard copy
- Online documentation
- Computer-based training
- Instructional
- System messages

Providing Guidance and Assistance

Useful guidance and assistance answers:

- What is this?
- What does it do?
- How do I make it do it?
- What is its role in the overall scheme of things?

Problems with Documentation

Problems:

- Organizational factors
- Time scale
- Theoretical rationale
- Resources

How Interact with Documentation

There are three broad:

- Finding relevant information
- Understanding what the documentation says
- Applying that understanding solve the problem

Principle

Principle:

- Instructions or Prompting
- Help Facility
- Contextual Help
- Task-Oriented Help
- Reference Help
- Wizards
- Hints or Tips

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ATAS SEGALA PERHATIAN
SEMOGA BERMANFAAT...

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